IMO 2026 — Problem 4

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2026-07-17

Source: `results/imo-2026/kimi-k3/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Approaches tried

- **Algebraic model of the cut.** A cut is a cevian: from a triangle with angles (A, B, C), cutting at the vertex with angle A with parameter $\alpha \in (0, A)$ (fully at Mulan's disposal) produces halves $\{B, \alpha, 180^\circ - B - \alpha\}$ and $\{C, A - \alpha, B + \alpha\}$ (see `lemmas/cut-geometry.md`). This reduces the game to pure angle arithmetic.
- **Retrograde analysis on a discretized game** (`code/retro.py`, `approaches/retrograde-analysis.md`): computed Mulan's winning region exactly for integer-degree states. At unit 1° she wins all states iff $\theta \in \{1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90\}^\circ$ — exactly the divisors of 180°; at unit 0.5° exactly $\{\theta : 180^\circ/\theta \in \mathbb{Z}\}$ again. This gave the conjecture.
- **Halving lemma** (`lemmas/halving-lemma.md`): for every θ, any triangle containing an angle $k\theta$ ($k \in \mathbb{Z}_{>0}$) is winning for Mulan (cut θ off $k\theta$, induct on k). So the whole question is whether Mulan can force a multiple of θ to survive a cut.

- **Mulan's side** (`approaches/marking-strategy.md`): if $\theta = 180^\circ/n$, a one-move "round-up" cut makes *both* halves contain multiples of θ (here $180^\circ = n\theta$ being a multiple of θ is essential), then the halving lemma finishes. Win in $\leq n-1$ moves.
- **Shan-Yu's side** (`approaches/invariant-avoidance.md`): if θ is not of the form $180^\circ/m$, Shan-Yu maintains the invariant "no angle is a multiple of θ" forever, starting from the equilateral triangle; the case analysis shows the invariant can only break when $180^\circ - k\theta$ is a multiple of θ, i.e. exactly when $\theta = 180^\circ/m$.
- Both strategies verified by exact rational-arithmetic simulation (`code/verify.py`): Mulan's strategy wins exhaustively with worst case exactly $n-1$ moves for all tested $n \leq 60$; Shan-Yu's invariant survived 200×300 random cuts for several $\theta \neq 180^\circ/m$.
- Dead ends: naive invariants for Shan-Yu such as "all angles $< \theta$" (works only for $\theta > 90^\circ$) and "no angle equal to θ" alone (not maintainable: Mulan can put θ in one half and nothing forbidden in the other); replaced by the multiple-of-θ invariant.

Current best

Answer: Mulan can guarantee victory in finitely many steps if and only if $\theta = \dfrac{180^\circ}{n}$ for some integer $n \geq 2$ (equivalently, iff $180^\circ/\theta$ is an integer). In that case she wins within $n-1$ moves: with the triangle's angles measured relative to θ, she cuts so that one angle is rounded up to the next multiple of θ; because $180^\circ = n\theta$ the supplementary angle at the cut is then also a multiple of θ, so *both* halves contain a multiple of θ; then she repeatedly cuts θ off the surviving multiple ($k\theta \to (k-1)\theta$) until an angle exactly θ appears in every branch. For all other θ, Shan-Yu starts with an equilateral triangle and always keeps a half having no angle equal to an integer multiple of θ (one exists among the two halves of any cut, precisely because $\theta \neq 180^\circ/m$), so Mulan can never win.

Full proof

Setup and angle bookkeeping

Throughout, describe a triangle by its multiset of angles (A, B, C), positive reals with $A+B+C = 180^\circ$. The game only depends on the angles.

Lemma 0 (the cut). *Suppose the current triangle has angles A at vertex X, B at Y, C at Z. Mulan's point P lies on some side; say it lies on YZ (the opposite vertex is then X). Put $\alpha = \angle YXP$. Then the two halves have angle multisets*

$$H_1 = \{B,\ \alpha,\ 180^\circ - B - \alpha\}, \qquad H_2 = \{C,\ A - \alpha,\ B + \alpha\},$$

and every value $\alpha \in (0, A)$ is achieved by a suitable interior point P of YZ.

Proof. See `lemmas/cut-geometry.md`. (Angle chase with $A + B + C = 180^\circ$; the range of α by continuity of $P \mapsto \angle YXP$ and the intermediate value theorem.) ■

So a move of Mulan consists of: choosing which angle of (A, B, C) is the *cut angle* A, choosing which of the remaining angles is B (the one that ends up in the same half as α), and choosing a real number $\alpha \in (0, A)$. Then Shan-Yu replaces the triangle by one of H_1, H_2.

Call an angle **marked** if it is an integer multiple of θ, i.e. equal to $k\theta$ for some integer $k \geq 1$ (necessarily $k\theta < 180^\circ$ since it is an angle of a triangle).

The halving lemma

Lemma 1. *Fix any θ. If the current triangle has a marked angle $k\theta$ ($k \geq 1$ integer), then Mulan can force victory from it within $k-1$ moves.*

Proof. Induction on k. If $k=1$ the triangle has an angle θ and the game is already over. If $k \geq 2$, let the angles be $A = k\theta$, B, C; Mulan cuts at A with $\alpha = \theta \in (0, A)$. By Lemma 0 the halves are

$$H_1 = \{B,\ \theta,\ 180^\circ - B - \theta\}, \qquad H_2 = \{C,\ (k-1)\theta,\ B+\theta\},$$

both genuine triangles since $B < 180^\circ - 2\theta$. If Shan-Yu keeps H_1, the game stops next (angle θ present); if he keeps H_2, Mulan wins from it within $k-2$ further moves by induction. Total $\leq k-1$ moves. ■

Theorem 1: $\theta = 180^\circ/n$ — Mulan wins

Theorem 1. *Let $\theta = 180^\circ/n$ for an integer $n \geq 2$. Then Mulan has a strategy that wins within $n-1$ moves from every initial triangle.*

Proof. If the initial triangle has a marked angle $k\theta$ ($1 \leq k \leq n-1$), Lemma 1 wins within $k-1 \leq n-2$ moves. So assume no angle of $\mathcal{T} = (A, B, C)$ is marked.

For an unmarked angle x define

$$d(x) = (\lfloor x/\theta \rfloor + 1)\theta - x \in (0, \theta),$$

the distance from x up to the nearest multiple of θ above it.

Claim 1. $d(A) + d(B) + d(C) \in \{\theta,\ 2\theta\}$.

Proof. Each $x + d(x)$ is a multiple of θ, and $(A + d(A)) + (B + d(B)) + (C + d(C)) = 180^\circ + d(A) + d(B) + d(C) = n\theta + d(A) + d(B) + d(C)$, so the sum of the d's is a multiple of θ; as each lies in $(0, \theta)$, the sum lies in $(0, 3\theta)$, hence equals θ or 2θ. ■

Claim 2. *There exist two distinct angles u, v of $\mathcal{T}$ with $d(u) < v$.*

Proof. Suppose not. Then in particular $d(A) \geq B$, $d(B) \geq C$, $d(C) \geq A$, and summing gives

$$d(A) + d(B) + d(C) \geq A + B + C = 180^\circ = n\theta.$$

By Claim 1 the left side is at most 2θ, so $n\theta \leq 2\theta$, i.e. $n = 2$, and all three inequalities must be equalities: $d(A) = B$, $d(B) = C$, $d(C) = A$. But then $A + d(A) = A + B$ is a multiple of θ by definition of $d(A)$, while $A + B = 180^\circ - C = 2\theta - C$ (here $n = 2$). Since $0 < A + B < 180^\circ$ we get $A + B = \theta$, hence $C = \theta$, contradicting that C is unmarked. ■

Mulan's move. Pick a pair $u \neq v$ with $d(u) < v$ (Claim 2) and let w be the third angle. Mulan cuts at the vertex with angle v, putting u in the role of B and

$$\alpha = d(u) \in (0, v).$$

Let $u + d(u) = k\theta$ ($k \geq 1$ integer, a multiple of θ just above u). Since $k\theta = u + \alpha < u + v < 180^\circ = n\theta$, we have $1 \leq k \leq n-1$. By Lemma 0 the halves are

$$H_1 = \{u,\ d(u),\ 180^\circ - u - d(u)\}, \qquad H_2 = \{w,\ v - d(u),\ u + d(u)\}.$$

In H_2 the angle $u + d(u) = k\theta$ is marked. In H_1 the angle

$$180^\circ - u - d(u) = n\theta - k\theta = (n-k)\theta$$

is marked as well, because $180^\circ = n\theta$ is a multiple of θ and $1 \le n-k \le n-1$. (This is the only place where the hypothesis $\theta = 180^\circ/n$ is used.)

So **both** halves contain a marked angle. Whichever half Shan-Yu keeps, the new triangle contains some marked angle $j\theta$ with $1 \le j \le n-1$, and by Lemma 1 Mulan wins from it within $j-1 \le n-2$ further moves.

Total: at most $1 + (n-2) = n-1$ moves. ■

Theorem 2: $\theta \neq 180^\circ/m$ — Shan-Yu avoids forever

Theorem 2. *Suppose $\theta \neq 180^\circ/m$ for every integer $m \ge 2$. Then Shan-Yu has a strategy (choice of initial triangle and of which half to discard) ensuring that no triangle in the game ever has an angle θ; in particular Mulan cannot guarantee a win.*

Proof. Shan-Yu maintains the invariant:

> **(I)** no angle of the current triangle is marked (i.e. no angle is an integer multiple of θ).

Since θ itself is a multiple of θ, invariant (I) implies the game never stops, so Mulan never wins.

Initial triangle. Shan-Yu builds the equilateral triangle $(60^\circ, 60^\circ, 60^\circ)$. If $60^\circ = k\theta$ for an integer $k \ge 1$, then $\theta = 60^\circ/k = 180^\circ/(3k)$, which is of the excluded form $180^\circ/m$ ($m = 3k \ge 3$). Hence (I) holds initially.

Maintenance. Assume $\mathcal{T} = (A, B, C)$ satisfies (I) and Mulan cuts at A with some $\alpha \in (0, A)$ and some choice of B, producing $H_1 = \{B, \alpha, 180^\circ - B - \alpha\}$ and $H_2 = \{C, A-\alpha, B+\alpha\}$ (Lemma 0). We show at least one half satisfies (I); Shan-Yu keeps it.

Case 1: $\alpha = k\theta$ is a multiple of θ. Then H_2 satisfies (I): C is unmarked by hypothesis; if $A - \alpha = j\theta$ for an integer $j \ge 1$ (note $A - \alpha > 0$), then $A = (j+k)\theta$ is marked, contradiction; if $B + \alpha = j\theta$ for an integer $j \ge 1$, then $B = (j-k)\theta$ with $j - k \ge 1$ (since $B > 0$), so B is marked, contradiction.

Case 2: α is not a multiple of θ, and $180^\circ - B - \alpha$ is not a multiple of θ. Then H_1 satisfies (I): B is unmarked by hypothesis, and the other two angles of H_1 are non-multiples by the case assumption.

Case 3: α is not a multiple of θ, but $180^\circ - B - \alpha = k\theta$ for some integer $k \ge 1$. (Note $0 < k\theta < 180^\circ$, since it is an angle of H_1.) Then H_2 satisfies (I):

- C is unmarked by hypothesis.
- $A - \alpha = A - (180^\circ - B - k\theta) = k\theta - C$. If $A - \alpha = j\theta$ for an integer $j \ge 1$ (it is positive), then $C = (k-j)\theta$ with $k - j \ge 1$ (since $C > 0$), so C is marked — contradiction.
- $B + \alpha = 180^\circ - k\theta$. If $B + \alpha = j\theta$ for an integer $j \ge 1$ (it is positive), then $180^\circ = (j+k)\theta$ with $j + k \ge 2$, i.e. $\theta = 180^\circ/(j+k)$ is of the excluded form — contradiction.

The three cases are exhaustive, so the invariant can always be maintained, and the proof is complete. ■

Conclusion

By Theorem 1, if $\theta = 180^\circ/n$ for an integer $n \geq 2$, Mulan guarantees victory within $n-1$ moves against any play of Shan-Yu (any initial triangle, any discards). By Theorem 2, for every other $\theta \in (0^\circ, 180^\circ)$ Shan-Yu prevents Mulan from ever winning. Hence

$$\boxed{\text{Mulan can guarantee victory} \quad \Longleftrightarrow \quad \theta = \frac{180^\circ}{n} \text{ for some integer } n \geq 2 \quad \Longleftrightarrow \quad \frac{180^\circ}{\theta} \in \mathbb{Z}.}$$

Explicitly: $\theta \in \{90^\circ, 60^\circ, 45^\circ, 36^\circ, 30^\circ, \frac{180^\circ}{7}, 22.5^\circ, 20^\circ, \ldots\}$.

Verification. The discretized game (1° and 0.5° grids, `code/retro.py`) yields exactly this family; the explicit strategies above were verified with exact rational arithmetic (`code/verify.py`): for $\theta = 180^\circ/n$ Mulan's strategy wins in exactly at most $n-1$ moves (worst case $= n-1$, tight), exhaustively over Shan-Yu's choices, for all tested $n \leq 90$ on random rational triangles; the pairing Claim 2 and the round-up move were additionally checked exhaustively on $\sim 6 \times 10^4$ unmarked grid triples for each $n \mid 360$, $n \leq 30$ (`code/check_pairing.py`); for $\theta \neq 180^\circ/m$ Shan-Yu's invariant survived 200×300 random cuts for each tested θ.